Sales Prediction Model

Real historical data, reproducible modeling, chronological validation and business interpretation.

1. Dataset and objective

Predict future monthly sales using historical sales and calendar-derived features. The original 176 observations are Australian wine sales, January 1980 to August 1994, from the Time Series Data Library / R forecast wineind series (distributed via pmdarima). Values are reported sales units; no currency is assumed.

The original dataset has only date and sales. Fifteen non-synthetic features are engineered: year, month, quarter, time index, cyclic month indicators, prior-month sales lags (1/2/3/6/12), past rolling averages (3/6/12) and a past rolling standard deviation.

2. Data preparation and validation

164 model-ready records: first 12 months used only as feature-history warm-up. 131 training records (Jan 1981–Nov 1991), 33 final test records (Dec 1991–Aug 1994). This is an approximately 80/20 split without shuffling. One-hot encoding and median imputation are fit inside training-only pipelines; no present-month target value is used in predictors.

3. Model performance on the untouched test interval

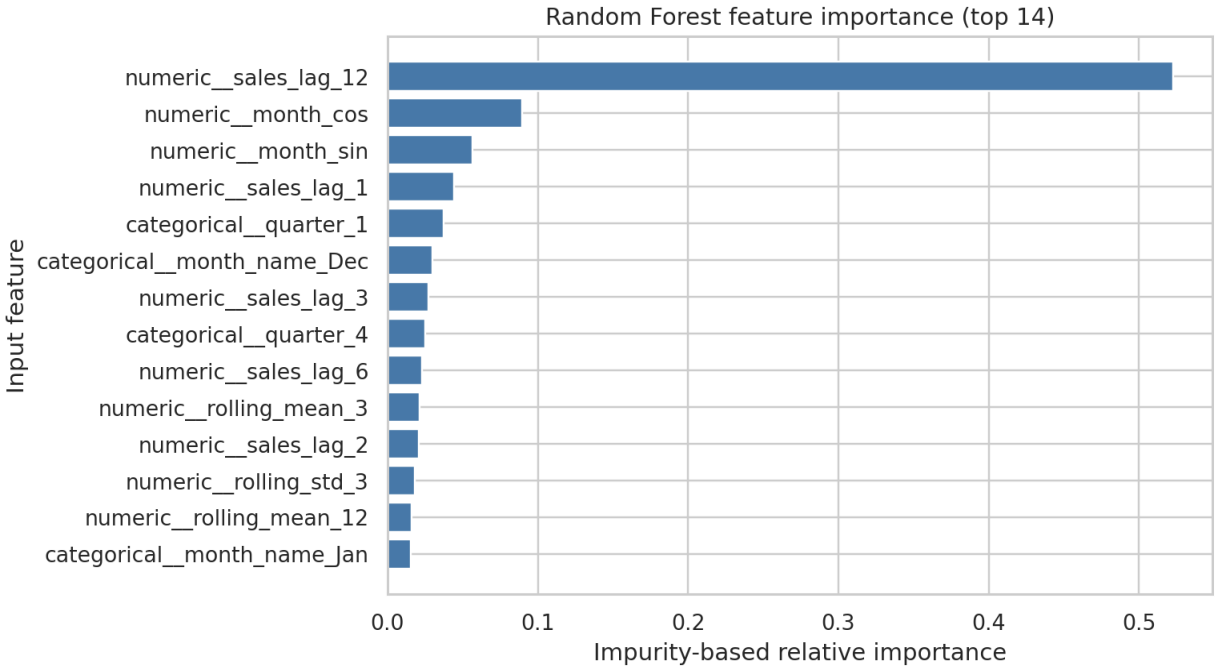
Candidate	CV RMSE	Test RMSE	Test MAE	Test R^2
Linear Regression	3,785	2,313	1,821	0.822
Random Forest	3,147	2,517	1,835	0.789
Tuned Random Forest	2,978	2,413	1,816	0.806
Seasonal naive (lag 12)	—	2,691	1,989	0.759

The training-only four-fold time-series CV favored Random Forest; 18 randomized hyperparameter candidates were searched. The tuned forest was preselected for forward forecasting without peeking at holdout results. On the final test, Linear Regression had the lowest RMSE/R^2, whereas tuned Random Forest had the lowest MAE.

RMSE emphasizes large errors; MAE measures absolute typical error; R^2 describes explained variability. Seasonal-naive lag-12 is a non-ML comparison, not an additional regression algorithm.

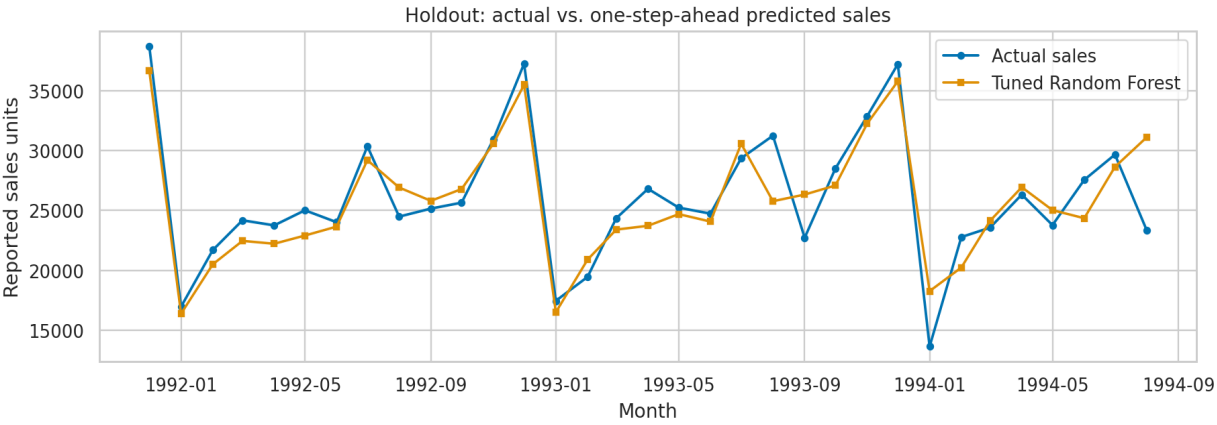
4. Random Forest feature importance

The strongest model input was numeric__sales_lag_12 (relative impurity importance 52.3%). Importances measure prediction dependence; they do not establish causal sales drivers.

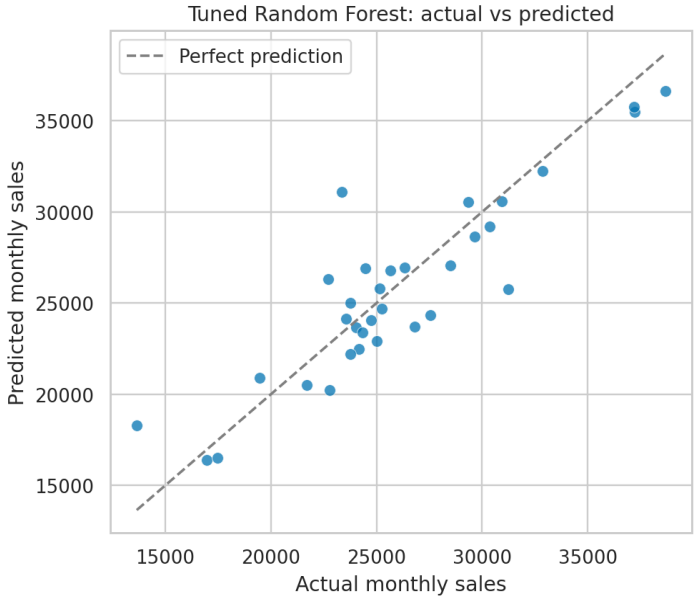


5. Actual vs. predicted: final holdout

The tuned model was selected using training cross-validation. The following historical holdout overlay represents sequential one-step-ahead predictions, not a single 33-month static forecast.



6. Actual vs. predicted scatter



7. Illustrative forward prediction

September 1994: 24,482.86 reported sales units. This prediction was generated by refitting the tuned model to all observed history through August 1994. The projected value is not verified ground truth.

8. Recommendations

Inventory: incorporate monthly seasonal patterns in replenishment, production and staffing plans, subject to later verification. Forecast operations: refresh each monthly prediction after actual previous-month sales are available. Monitoring: track error by month and compare each candidate with a simple seasonal benchmark.

9. Limitations and validity

The sales history is short and old (1980–1994), and lacks prices, promotions or other independently sourced covariates. Lag and rolling predictors require real recent sales as inputs. In holdout scoring, previous months of the holdout have already been observed before predicting the next month: reported metrics therefore apply to a one-month rolling forecast and do not validate a 33-month unsupported forecast.

Tree feature importance is associational rather than causal. Economic shocks, changing consumer behavior and missing exogenous factors can invalidate generalization. Source units are not labeled as a currency. This is a clearly documented educational prototype, not a current operational forecast.

Source: https://pkg.robjhyndman.com/forecast/reference/wineind.html